Determinants of Intention to Use Artificial Intelligence-based Diagnosis Support System Among Prospective Physicians

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Abstract

This study aimed to develop a theoretical model to explore the behavioral intentions of medical students to adopt an AI-based Diagnosis Support System. This online cross-sectional survey used the unified theory of user acceptance of technology (UTAUT) to examine the intentions to use an AI-based Diagnosis Support System in 211 undergraduate medical students in Vietnam. Partial least squares (PLS) structural equational modeling was employed to assess the relationship between latent constructs. Effort expectancy (beta = 0.201, $p < 0.05$) and social influence (beta = 0.574, $p < 0.05$) were positively associated with initial trust, while no association was found between performance expectancy and initial trust ($p < 0.05$). Only social influence (beta = 0.527, $p < 0.05$) was positively related to the behavioral intention. Conclusions: This study highlights positive behavioral intentions in using an AI-based diagnosis support system among prospective Vietnamese physicians, as well as the effect of social influence on this choice. The development of AI-based competent curricula should be considered when reforming medical education in Vietnam.